আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম

الجامعة الإسلامية العالمية شيتاغونغ

International Islamic University Chittagong

LAB REPORT- 01

COURSE CODE: CSE-2340

COURSE TITLE: SOFTWARE DEVELOPMENT 1

LAB TOPIC:

JAVA PRACTICE PROBLEM

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DD-MM-YYYY

Marks:

Signature:

SET-01 (Beginner)

Problem-01:

Java Intro

Input-Output Screenshot:

```
E:\C231045\java>javac JavaIntro.java

E:\C231045\java>java JavaIntro
Hello World
```

Approach:

I have recently begun learning the Java programming language. After installing the Java Development Kit (JDK) on my system, I successfully wrote and executed my first "Hello, World!" program, which displays the output "Hello Programmer" in the terminal.

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SET-01 (Beginner)

Problem-02:
Java Get Started

Input-Output Screenshot:

```
E:\C231045\java>javac GetStarted.java

E:\C231045\java>java GetStarted
welcome the new world

E:\C231045\java>_
```

Approach:
It is a get started program where I have first defined the filename as GetStarted.java and the first outer class is named also GetStarted to match the filename which is compulsory in Java. then inside we must have to include the main method which is a starting point of a Java program. Inside that main method I have written the text printed into the terminal which is “Welcome to the new world”.

SET-01 (Beginner)

Problem-03:
Java Syntax

Input-Output Screenshot:

```
E:\C231045\java>javac StudentInfo.java

E:\C231045\java>java StudentInfo
Student Information:
Name: ABDUR RAHMAN
ID: C231045
Semester: 3RD

E:\C231045\java>
```

Approach:
I now know how to write simple Java programs from past program. So, in this Syntax.java program I have written a program that contains multiple statements printing my Id, Name , SEMESTER, Those statements are in the main method.

SET-01 (Beginner)

Problem-04:
Java Output

Input-Output Screenshot

```
E:\C231045\java>javac OutputDemo.java

E:\C231045\java>java OutputDemo
Name: ABDUR RAHMAN | ID: C231045 | Semester: 3RD

Arithmetic Operations:
10 / 4 = 2.5
10 * 4 = 40

E:\C231045\java>
```

Approach:
Here, I have written a Java program using both println and print method of Java. The difference between them is println automatically takes a line break where print method doesn't. And then I have written my name id and 2 statements that do some arithmetic operations like divisions and multiplication. Though, while doing the division I have to explicitly mark one of the operand a floating point, otherwise we won't be able to see the remainder.

SET-01 (Beginner)

Problem-05:
Java Comments

Input-Output Screenshot

```
E:\C231045\java>javac JavaComments.java

E:\C231045\java>java JavaComments
Learning about Java Comments!
Comments help document your code
Documentation comments generate API docs
Value of x: 10

E:\C231045\java>
```

```java
public class JavaComments {
    public static void main(String[] args) {
        /* Display a message about Java comments */
        System.out.println("Learning about Java Comments!");

        /*
           Multi-line comment example:
           This comment spans multiple lines
           It can contain detailed explanations
           All text here is ignored by the compiler
        */
        System.out.println("Comments help document your code");
```

Approach:
I Create a Java class named `JavaComments` with the main method as the program entry point. Use only multi-line comments (`/* */`) to explain code functionality, demonstrate commented-out code, and describe variable declarations. These comments are ignored by the compiler and serve only for documentation. Compile using `javac JavaComments.java` and run using `java JavaComments` to display the output.

SET-01 (Beginner)

Problem-06:
Java Variables

Input-Output Screenshot

```
E:\C231045\java>javac JavaVariables.java

E:\C231045\java>java JavaVariables
Full Name: Abdur Rahman
Sum of a and b: 40
Birth Year: 1995

E:\C231045\java>_
```

Approach:
I creat a Java class named JavaVariables with the main method as the program entry point. Declare three String variables where firstName and lastName store individual name components, then concatenate them into fullName using the plus operator and print it. Declare two integer variables a and b, perform arithmetic addition, and print their sum. Declare a constant integer variable using the final keyword to store birthYear, ensuring its value cannot be modified throughout the program, then print it. Compile using javac JavaVariables.java and execute using java JavaVariables to display all outputs.

SET-01 (Beginner)

Problem-07:
Java Data Types

Input-Output Screenshot

```
E:\C231045\java>javac JavaDataTypes.java

E:\C231045\java>java JavaDataTypes
10 5.5 A true

E:\C231045\java>
```

Approach:
I write a Java class named JavaDataTypes that demonstrates four primitive data types. Declare an integer variable a to store a whole number value of 10. Declare a double variable b to store a decimal value of 5.5. Declare a char variable c to store a single character 'A'. Declare a boolean variable d to store a logical value true. In the main method, use System.out.println to print all four variables together by concatenating them with spaces. Compile using javac JavaDataTypes.java and run using java JavaDataTypes to display the output showing all four data type values in a single line.

SET-01 (Beginner)

Problem-08:
Java Type Casting

Input-Output Screenshot

```
E:\C231045\java>javac TypeCasting.java

E:\C231045\java>java TypeCasting
2.78
2
```

Approach:
TypeCasting with the main method as the program entry point. Declare a double variable num and assign it a decimal value of 2.78. Perform explicit type casting by converting the double value to an integer using (int) before the variable ,demonstrating how explicit casting removes fractional parts when converting from double to int.

SET-01 (Beginner)

Problem-09:
Java Operators

Input-Output Screenshot

```
E:\C231045\java>javac OperatorsExample.java

E:\C231045\java>java OperatorsExample
15
5
50
2

E:\C231045\java>
```

Approach:
Here in the named OperatorsExample with the main method. Declare two integer variables a and b with values 10 and 5. Perform arithmetic operations including addition, subtraction, multiplication, and division. Print each result sequentially using System.out.println. Compile and run to display all arithmetic outputs.

SET-01 (Beginner)

Problem-10:

Java Strings

Input-Output Screenshot

```
E:\C231045\java>javac StringExample.java

E:\C231045\java>java StringExample
25
JAVA PROGRAMMING CSE IIUC

E:\C231045\java>
```

Approach:

The program creates a class StringExample with a main method. Inside, a string text is initialized with "Java Programming cse iiuc". The length() method counts all characters (including spaces) and prints 26, while the toUpperCase() method converts the entire string to uppercase and prints "JAVA PROGRAMMING CSE IIUC". Both methods operate on the original string without modifying it, as strings in Java are immutable.

SET-01 (Beginner)

Problem-11:

Java Math

Input-Output Screenshot

```
E:\C231045\java>javac Maths.java

E:\C231045\java>java Maths
checking max between 5 and 6 : 6
checking min between 7 and 2 : 2
2^3 = 8.0
square root of 4 is 2.0
abs value of -4.7 is 4.7
round of 5.6 is 6
ceil point 5.6 is 6.0
floor point of 5.6 is 5.0
random number between 0 - 100 : 64

E:\C231045\java>
```

Approach:

The program demonstrates Java's Math class methods including max(), min(), pow(), sqrt(), abs(), round(), ceil(), floor(), and random() to perform various mathematical operations and display the results.

SET-01 (Beginner)

Problem-12:

Java Booleans

Input-Output Screenshot

```
E:\C231045\java>javac BooleanExample.java && java BooleanExample
true

E:\C231045\java>
```

Approach:

the program creates a class BooleanExample with a main method. Inside, a boolean variable isJavaFun is declared and initialized with the value true. The System.out.println() method then prints the boolean value directly to the console. Boolean in Java can only hold two values: true or false, and when printed, it displays the literal value without quotes.

SET-01 (Beginner)

Problem-13:

Java If...Else

Input-Output Screenshot

```
E:\C231045\java>javac IfElseExample.java

E:\C231045\java>java IfElseExample
Positive

E:\C231045\java>
```

Approach:

The program creates a class IfElseExample with a main method. An integer variable number is initialized with 231045. The if-else statement checks whether the number is greater than zero. Since 231045 is positive, the condition number > 0 evaluates to true, so the code inside the if block executes and prints "Positive". If the number were negative or zero, the else block would execute and print "Negative". This demonstrates basic conditional branching in Java.

SET-01 (Beginner)

Problem-14:

Java Switch

Input-Output Screenshot

```
E:\C231045\java>javac SwitchExample.java

E:\C231045\java>java SwitchExample
Tuesday

E:\C231045\java>
```

Approach:

First, an integer variable named day is declared and initialized with a value of 3. Then, the switch statement evaluates this variable and compares it against multiple predefined case labels. When the value matches case 3, the program executes the corresponding print statement to display "Tuesday

SET-01 (Beginner)

Problem-15:

Java While Loop

Input-Output Screenshot

```
E:\C231045\java>javac WhileLoop.java

E:\C231045\java>java WhileLoop
1
2
3
4
5

E:\C231045\java>
```

Approach:

this program is to demonstrate the use of a while loop for repetitive execution in Java. First, a counter variable i is initialized to 1. Then, the while loop checks the condition i <= 5 before each iteration. As long as the condition evaluates to true, the loop executes the block of code that prints the current value of i and then increments it by 1 using i++. This process repeats until i becomes 6, at which point the condition i <= 5 becomes false and the loop terminates. This approach effectively prints numbers from 1 to 5 in sequence, showcasing how a while loop can be used when the number of iterations is determined by a condition rather than a fixed count.

SET-01 (Beginner)

Problem-16

Java For Loop

Input-Output Screenshot

```
E:\C231045\java>javac ForLoop.java

E:\C231045\java>java ForLoop
1
2
3
4
5

E:\C231045\java>
```

Approach:

This program i uses a for loop to print numbers 1 to 5. The loop initializes i to 1, runs while i <= 5, and increments i after each iteration, printing the value each time until the condition fails.

SET-01 (Beginner)

Problem-17:

Java Break/Continue

Input-Output Screenshot

```
E:\C231045\java>javac BreakContinue.java

E:\C231045\java>java BreakContinue
1
2
4

E:\C231045\java>
```

Approach:

This program uses a `for` loop from 1 to 5. When `i` is 3, `continue` skips printing. When `i` is 5, `break` exits the loop. Thus, only 1, 2, and 4 are printed.

SET-01 (Beginner)

Problem-18:

Java Arrays

Input-Output Screenshot

```
E:\C231045\java>javac ArrayExample.java

E:\C231045\java>java ArrayExample
10
20
30

E:\C231045\java>
```

Approach:

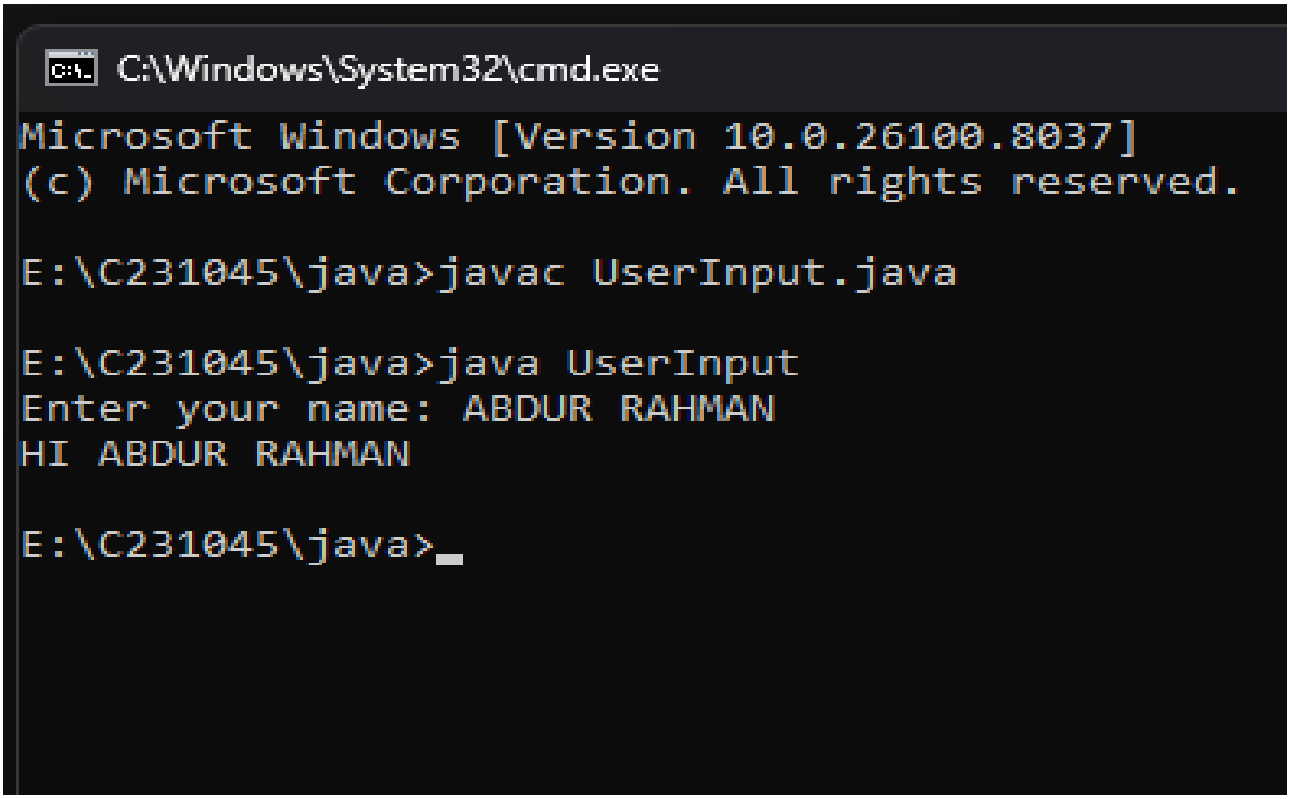
I creat program declares an integer array `numbers` with values 10, 20, and 30. A `for` loop iterates from index `0` to `numbers.length - 1`, printing each element sequentially. Thus, output is 10, 20, and 30 each on a new line.

SET-01 (Beginner)

Problem-19:

Java User Input

Input-Output Screenshot



Approach:

my program imports the Scanner class to accept user input. It creates a Scanner object to read from the keyboard, prompts the user to enter their name, stores the input in a String variable, and then displays a greeting message "HI" followed by the entered n

THANK YOU